



Implications of the power sector emissions on the deep decarbonisation of Nigeria: evaluating long-term model based scenarios

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Abstract

Limited energy access is one of Nigeria's most pressing development challenges. At the same time, Nigeria has pledged to attain net-zero by 2060. However, science-based evidence that is necessary to support this ambition is still lacking. Thus, this work investigates the implications of the power sector's emission on Nigeria's net-zero goal. This research employs three scenarios to decarbonise the Nigeria's power sector up to 2060. The three scenarios—Current Policy Scenario (CPS), Gas Economy Scenario (GES), and Renewable Energy Scenario (RES), were modelled using the Low Emission Analysis Platform, on the premise that Nigeria achieves its energy-for-all agenda by 2030. The findings suggest that the Nigerian power capacities must grow to 398 GW, 250 GW, and 347 GW in the CPS, GES, and RES, by 2060. Also, the decarbonisation scenarios suggest that the power sector's emissions would be reduced to 0.5, 0.1, and 0.1 MMT CO_{2-eq}, respectively, for the CPS, GES, and RES, as against 112.5 MMT CO_{2-eq} in the baseline scenario, by 2060. Key message to policy makers is that Nigeria can fulfil its energy-for-all obligation while simultaneously decarbonising the electricity sector towards attaining its net-zero goal. However, this would require a significant rollout of solar energy (in either the CPS or RES scenarios) or large-scale carbon capture installations (in GES). Therefore, unless the government urgently prioritises an uptake of clean technologies, and creates a political environment that encourages investment in clean energy, the country is likely to fall behind in its climate and energy access goals.

Highlights

- Three emission reduction scenarios, namely, current policy, gas economy and renewable energy scenarios were proposed for Nigeria.
- The scenarios suggested that it is possible to reduce the power sector emissions by 2060 from 112.5 to 0.1–0.5 MMT CO_{2-eq}, without compromising universal energy access.
- Significant rollout of solar is required in the renewable energy-based scenario, while carbon capture technology is required in the gas-based scenario.

Keywords Net-zero · Energy transition · Scenario development · Energy modelling · Renewable energy technology · Climate change

Nomenclature

Abbreviations

AL	Activity level
CPS	Current policy scenario
EI	Energy intensity
ETP	Energy transition plan
GES	Gas economy scenario
LEAP	Low Emission Analysis Platform
MMT	Million metric tonne
NDC	Nationally Determined Contribution
PRM	Planning reserve margin
RES	Renewable energy scenario
RM	Reserve margin

Notations/Symbols

C_{Endo}	Endogenous capacity
C_f	Capacity factor
CO_{2-eq}	Carbon dioxide equivalent
D_p	Peak demand
E_d	Energy requirement
ef_t	Emission factor
L_f	Load factor
P_t	Power output
I	Sector
T	Technology

Units

%	Percentage
-	Dimensionless unit
GWh	Gigawatt hour
kWh/cap	Kilowatt hour per capita
MMT CO_{2-eq}	Million metric tonne carbon dioxide equivalent
MW	Megawatt
PJ	Petajoule
tonne/GWh	Tonne per gigawatt hour

1 Introduction

It is now widely believed that the goal to limit global average temperature rise to below 2 °C (above pre-industrial levels) and to pursue the more ambitious 1.5 °C (UNFCCC 2015) calls for the deep decarbonisation of the global economy. Equally commonly recognized is that the 1.5 °C target equates to net-zero carbon emission, implying that carbon removal is expected to match anthropogenic carbon emissions.

Nigeria, being one of the signatories to the Paris Agreement, has taken commendable steps to decarbonise its economy by committing to a net-zero target by 2060 in the declaration at the 26th Conference of the Parties to the climate agreement (COP26) held in Glasgow, UK. Nigeria also has an ambitious Nationally Determined Contribution (NDC) that targets 20% and 47% emission reduction from the business as usual on unconditional and conditional bases, respectively. The country also has an ambitious Energy Transition Plan (ETP) on which basis the Glasgow net zero announcement was made. However, with the highest unelectrified population globally – 90 million out of Nigeria’s more than 200 million population are without access to electricity (REAN 2024) – Nigeria will need to massively grow its power capacity to meet its goal of universal energy access and simultaneously keep down the carbon emissions coming from the power sector.

Concerns are pervasive that Nigeria’s ambitious NDC and the net-zero goals are simply mere declaration that are not yet backed with evidence-based policy and strategic pathways necessary for their implementation (Adeyanju et al. 2020; Nwokolo et al. 2023; Owebor et al. 2025a, b). Nwozor et al. (2021) has advanced the argument that one of the root causes of non-actualization of green energy transition targets in Nigeria is the lack of policies and projections that are supported by sustainable models and pathways. A cursory look into the Nigerian economy suggests that transport, oil and gas, industry, power, building, agriculture, forestry and other land use (AFOLU) are all critical sectors that must be targeted for deep decarbonisation in a net-zero roadmap.

In the hierarchy, the power sector is not the major culprit of greenhouse gases in Nigeria, but it is unarguably the sector that is most viable to drive a robust decarbonization, since most renewable energy need to be converted into electricity before use (Dioha and Kumar 2020). Apart from decarbonising its own GHGs emissions, studies and experience have proven that electrification is key in decarbonising other sectors. Key aspects include the electrification of transport system (via electric vehicles), electrification of industrial heating processes (electricity-based steam and heat production), electrification of the building and residential sector (electricity-based space heating in buildings), amongst others (Fortes et al. 2019). To this end, scenario modeling has played a major role in guiding decarbonisation efforts (Emodi et al. 2022; Omene-Tietie et al. 2021). Scenario affords both researchers and policy makers the opportunity to explore different future trajectories and to analyze the consequences of these alternative futures at different time horizon (short, mid, and long term), to inform decisions. Scenarios are not tools that can be used to predict the future. Scenario development projects into the future based on existing and historical data, as well as making reasonable assumptions about what the future could look like.

The first of its kind, this research uses three unique scenarios to explore different energy mix alongside the carbon emissions implications of these scenarios on the net-zero decarbonisation of the Nigerian power sector up to 2060. The three emission reduction scenarios - Current Policy Scenario (CPS), Gas Economy Scenario (GES), and Renewable Energy Scenario (RES) were framed and modelled using the Low Emission Analysis Platform (LEAP) with the assumption that Nigeria achieves universal energy access by 2030. Assumptions were also made on the penetration of power technologies, transmission and distribution, and the demand for electrical energy, for each scenario.

Although there are existing studies on the decarbonisation of the Nigerian economy, no single paper has carried out a deep dive into the decarbonisation of the Nigeria power sector with the country’s net-zero vision in focus and a time horizon that stretches to 2060. This

research aims at developing pathways through which the Nigerian power sector can contribute to the net-zero vision of the country. To achieve this aim, this research answers the following fundamental questions (i) what is the power requirement to drive a deeply decarbonised Nigerian economy by 2060, while meeting the universal energy access (Agenda 2030)? (ii) what is the emission profile of the power sector in a deeply decarbonised Nigerian economy by 2060? (iii) what is the emission factor and electricity per capita in the decarbonisation scenarios? To answer these questions, this study explores different scenarios that imagine: (i) the utilization of the Nigeria's gas reserves (ii) the utilization of the abundant renewable energy resources in Nigeria, especially, solar (iii) the harnessing of the strengths of different emission reduction policies in Nigeria, notably, the NDC and the ETP.

1.1 Background/national circumstances

Nigeria occupies an important position in the geopolitics and Africa's climate landscape. Hence, what happens in the country could have implications in how the rest of the sub-region responds to climate change and the emergent global energy transition. Nigeria has the fourth largest economy (IMF 2024) and it is the most populous country in Africa. Rich in oil and gas, Nigeria supplies gas-produced electricity to several neighboring countries including Chad, Niger and Ghana. Nigeria is heavily dependent on fossil fuel with about 86% of foreign exchange earning coming from oil and gas. Coming behind South Africa, Egypt and Algeria, Nigeria has the fourth highest GHG emission in Africa (IEA 2021). At the same time, Nigeria is very vulnerable to climate change. According to the Notre Dame Global Adaptation Initiative (ND-GAIN), Nigeria is ranked 154th out of 185 countries on its country index, making it among the 30 most vulnerable countries in the world (ND-GAIN 2021).

Like many other Sub-Saharan African countries, Nigeria exhibits an unfortunate energy-access paradox. Despite having abundant renewable and non-renewable energy sources, about 86–90 million out of Nigeria's more than 200 million population, mostly in the rural area, remain unconnected to the grid, marking Nigeria as having the largest un-electrified population globally (Husein et al. 2024; Owebor et al. 2025a; REAN 2024). While the present power demand is estimated at 17,520 MW, the total generation capacity is at 6,107 MW. To close this gap, Nigeria plans to develop 30,000 MW of new installed capacity by 2030. Electricity generation in Nigeria is composed mainly of gas thermal power sources, (72.73%), hydropower sources, (12.82%), Diesel/HFO sources (14.17%) and renewable sources (0.28%). Solar contributes a mere 0.2% of Nigeria's installed capacity, making its contribution to the country's energy mix almost insignificant (REAN 2024). Meanwhile, Nigeria boasts significant solar potential, with daily irradiation equivalent to energy from over a million tons of oil, far surpassing its oil and gas output (Owebor et al. 2021).

2 Literature review

Emissions reduction pathways have been proposed for Nigeria in the literature. In a study, Emodi et al. (2017) proposed four energy policy scenarios for Nigeria to mitigate its GHG emissions by 2040. The scenarios included business as usual (BAU), low-carbon moderate scenario (LCM), low-carbon advanced scenario (LCA), and green optimistic scenario (GO). In the study, they proposed that energy demand will decrease from 3075 PJ in a the BAU to

2941 PJ, 2488 PJ, and 2249 PJ, in the LCM, LCA, and GO, respectively, by 2040. They also showed that GHG emissions can decrease by 38.3%.

Ibrahim and Kirkil (2018) modelled energy conservation (EC) and renewable energy (REN) scenarios to demonstrate that Nigeria can attain uninterrupted power supply and reduce its GHG emissions by 2040. A key aspect of the study was the proposal of 100% and 50% efficient lighting and efficient energy intensive technologies, respectively by 2040. The study showed that GHG emissions would be 84.6 and 1.02 MMT CO_{2-eq} in the EC and REN, respectively, by 2040. Their findings suggested that the EC was more realistic given the cost of renewable energy technologies at the time. While it is commendable that the REN approaches net-zero from 2040, a gradual adoption of energy efficient technologies would have been more plausible.

Yetano-Roche et al. (2020) examined three scenarios for the Nigeria power sector by 2030, drawing inspiration from the universal energy access. The scenarios included BAU, green transition and moderate change scenarios. Their findings suggested that the green transition achieves 90% energy access by 2030, as compared to 53.3 and 65.9%, respectively, in the BAU and the moderate change scenarios. Also, they suggested that the grid connected access would attain Tier 3 level of energy access, which equates to 500 kWh per capita, annually, whereas off-grid access would attain Tier 2 level of energy access at 100 kWh per capita. However, their findings fail to meet the universal energy access and inadequately addressed the country's decarbonization concern as the power sector emissions in all their three scenarios were within 146–153 MMT CO_{2-eq} by 2030.

Bamisile et al. (2020) investigated different scenarios for Nigeria, with 100% electrification target by 2030, in response to the country's vision 2030. They used the EnergyPLAN model in carrying out the study. They suggested that electricity demand would increase to 200 TWh by 2030. Furthermore, they suggested that as a single source, 36,000 MW of natural gas can meet the electricity demand by 2030. Nevertheless, the study failed to show the environmental footprint of the different scenarios.

Ismail et al. (2020) developed three scenarios, namely, mild renewable energy scenario, extreme renewable energy scenario and total green energy scenario, in addition to the BAU scenario. They used the LEAP modelling architecture in their study. From their findings, it was not clear if universal energy access or at what year that universal energy access, would be attained. Nevertheless, the study suggested that power generation capacity would be 71,100 MW in the total green scenario by 2050 as against 21,900 MW in the BAU scenario by 2050. It was also suggested that total GHG emissions would be between 119.3 and 231 MMT CO_{2-eq} by 2050, suggesting that net-zero target was not feasible.

To demonstrate how Nigeria can actualize the Paris mandate, Dioha and Kumar (2020) examined four scenarios. The scenarios included unconditional NDC, conditional 1 NDC, conditional 2 NDC, and emission tax scenarios. The scenarios inherited the assumptions that were made in the conditional and unconditional NDC of the Federal Republic of Nigeria (FGN 2015), as well as, assume different levels of international support with respect to funding and technology transfer, and the impact of carbon pricing. Their findings suggested that the residential sector has the highest potential to reduce energy demand, whereas, the power sector has the greatest potential to mitigate carbon dioxide emissions (CO₂). Furthermore, they proposed that nuclear energy will play a key role in the Nigeria power sector, by contributing 81.5% to the country's energy mix by 2060. While it has been admitted

that nuclear energy is very significant for baseload, the dominance of nuclear over solar, by 2060, is an issue that needs to be reconsidered.

Kim et al. (2021) also modelled two scenarios for optimal electricity generation in Nigeria by 2060. They used the Model for Energy Supply Strategy Alternatives and their General Environmental Impact (MESSAGE) and the Simplified Approach to Estimating Electricity Generation's External Costs and Impacts (SIMPACTS) tools in modeling both scenarios. Their first scenario was based on the premise that there is no constraint on CO₂ emissions, whereas, in the second scenario, some constraints were imposed on CO₂ emissions. Their results showed that gas power plants would contribute 98.3% to the power mix in the no carbon dioxide constraint scenario. Whereas, similar to the work of Dioha and Kumar (2020), nuclear would contribute more than 80% to the power mix of the carbon dioxide constrained scenario.

From the foregoing, it could be seen that the existing studies are limited in one or more of the following (i) universal energy access (ii) emission reduction capacity (iii) net-zero target (iv) energy efficiency target (v) renewable energy capacity (vi) overreliance on nuclear energy. More also, some of these studies were carried out on either short-term (2030) or mid-term (2040) basis. It is against this backdrop that this research was conducted. Therefore, this work investigates a long-term net-zero power sector modelling using the LEAP framework to address the research questions outlined in the introductory section.

3 Methodology

In this study, three policy (emissions reduction) scenarios were developed. This is in addition to the baseline scenario. The emissions reduction scenarios were modelled with the Nigeria's net-zero vision in focus. The Gas Economy Scenario (GES) assumes that Nigeria will explore its huge gas reserve to power its economy, alongside decent penetration of renewable energy. The Current Policy Scenario (CPS) assumes that Nigeria implements her Energy Transition Plan (ETP) and Nationally Determined Contribution. In the Renewable Energy Scenario (RES), Nigeria explores and deploys its renewable energy resources more quickly, and electrifies most of the energy sectors. All scenarios, except the baseline, assume that Nigeria meets its universal energy access goal by 2030. The timeframe of the scenarios stretches up to 2060. The base year of the scenario is 2018 – this was chosen because it was the latest year with good data availability. The scenarios were modelled and implemented in the Low Emission Analysis Platform (LEAP), Version 2020.1.0.81 (Heaps 2021). Figure 1 shows a simplified view of the methodology.

3.1 The Nigerian LEAP model

LEAP (Low Emission Analysis Platform) is a bottom-up (BU) energy modelling tool that is used for the quantitative analysis of energy systems, pollutant emissions of energy and non-energy related sources. Also, it is very useful in cost and benefits analysis. The Nigeria LEAP model that was developed in this research comprised of electrical energy demand, and energy (power only) transformation. The demand-side of the Nigeria LEAP modelled the energy demand in residential, services, transport and industry sectors. The transformation modelled electricity generation (grid and off-grid) and transmission of grid electricity.

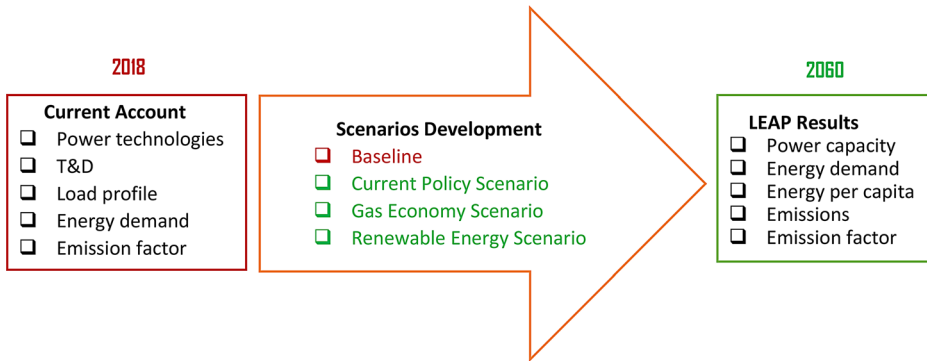


Fig. 1 Simplified view of the methodology

The emission reduction scenarios, alongside the baseline were fed into the Nigerian LEAP model to obtain the Nigerian power sector's emission profile under different policy assumptions, the power requirements of the different scenarios and the power technology mixes. The Nigerian LEAP model took into considerations some macroeconomic indicators such as GDP, and GDP per capita. Some other assumptions which had strong influence on the energy requirements in the different sectors of the economy and fed into the Nigerian LEAP model included, population, population growth rate, and rate of urbanisation. Notwithstanding, in this research, attention was not given to the emissions from other sectors of the economy, apart from the power sector.

The greenhouse gas emissions of power generation are calculated using Eq. (1)

$$GHG = \sum_t ef_t P_t \quad (1)$$

where, P_t (GWh) and ef_t (tonne/GWh) are the power output of power technology, t , and emission factor association with power generation (of technology, t), respectively.

In the LEAP model, the capacities of power technologies are entered exogenously, or calculated endogenously. The endogenous calculation (internal within LEAP) is adopted from Awopone et al. (2017) and presented in Eq. (2)

$$C_{Endo} = D_p (PRM - RM) \quad (2)$$

where, C_{Endo} (MW), PRM (%), RM (%), and D_p (MW), are endogenous capacity, planning reserve margin, reserve margin before addition, and peak requirement, respectively. The reserve margin and peak requirement are determined using Eq. (3) and Eq. (4), respectively.

$$RM = \left(\frac{C_m - D_p}{D_p} \right) \times 100 \quad (3)$$

$$D_p = \frac{E_d}{8760L_f} \quad (4)$$

where, C_m (MW), E_d (MWh), and L_f (-), are module capacity, energy requirement, and load factor (calculated internally as the average height of the load curve), respectively.

$$C_m = (C_e + C_0)C_f \quad (5)$$

where, C_e and C_0 are exogenous capacity and endogenous capacity added previously, respectively. C_f is the capacity factor used to adjust the intermittent renewable energy. Its value is 100%, 10–30%, 10–40%, 60–100% and 80–100% for thermal, solar, wind, hydro, and biomass, respectively (Heaps 2021).

The total electrical energy demand across each scenario is calculated in Eq. (6)

$$TC = \sum_i \sum_t (AL_{i,t} EI_{i,t}) \quad (6)$$

where, AL (-), EI are activity level and energy intensity of end-use technology, respectively; i , and t are sector and technology, respectively.

The Nigeria grid power comprised of large hydro and gas thermal power plants. To model the power generation system in the Nigeria LEAP, the hydro power plants were aggregated as 1384 MW (base year value), while the gas thermal power plants were aggregated as 5845 MW (base year value). The operational characteristics of the plants included the power capacity, life of plant, efficiency of plant, maximum availability. Similarly, the off-grid power was aggregated as 25,051 MW (base year value), with most of the contributions coming from fossil fuel generators (at 99.8%), see Table 1 for more information.

Table 1 Key baseline assumptions

Key Parameters	Assumption		
Base year	2018		
Base year population	195.88 million people		
Population growth rate	2030 = 2.5%, 2040 = 2.3%, 2050 = 2.0%, 2060 = 1.7%		
Urbanization rate	Base year – 43.5%, 2060–76.4%		
Electrification rate	56.5%		
Base year GDP	649.13 billion USD		
Annual GDP Growth Rate	2020 = -1.79%, 2021 = 3.64%, 2022 = 6.2%, 2023 = 5.8%, 2024 = 5.5%, 2025 = 5.2%, 2030 = 5.0%, 2040 = 4.7%, 2050 = 4.5%, 2060 = 5.0%.		
Base year power generation data	Grid generation (7,228 MW): Large Hydro Plants = 1383 MW, Gas Turbines = 5,845 MW; T&D losses = 17% Off-grid generation (25,056 MW): Standalone Solar PV (20 to 200 W) = 1 MW, Rooftop Solar PV (200 W to 20 KW) = 10 MW, Small Hydro = 45 MW, Fossil fuel based self-generation = 25,000 MW		
Household electrification	Lighting	Cooking	Other electricity use
	Grid = 44.8%, off-grid = 31.2% of households.	1.3% of households.	AC = 2467.7GWh, Fans = 76.7 GWh, TVs = 191GWh, Freezers = 557GWh.
Transport electrification	Electric vehicles = 0% of vehicles		
Services (electricity use)	27.8% of commercial energy consumption		
Industry (electricity use)	10.82% of energy consumption		

Source: Federal Government of Nigeria (FGN 2023); Okereke et al. (2024)

3.2 Scenarios and main assumptions

The decarbonisation of the Nigeria power sector with the Nigeria net-zero target in focus presents an opportunity to deliver on long standing issues such as universal energy access and vision 2020:20 (Nigeria to be listed among the top twenty economies of the world). Several potential development strategies are available. However, three emissions reduction scenarios have been modelled in the following sub-sections, and non-policy related assumptions such as population, GDP, were considered to be the same across all the scenarios.

3.2.1 Baseline scenario

The baseline scenario was based on the premise that the energy access in Nigeria would not improve substantially, and that the present rate of development would be maintained. In this scenario, the major drivers of electricity consumption were population growth rate, urbanization, and GDP. The baseline scenario takes into consideration the following exogenous assumptions:

- i. No restriction on fossil fuel usage for power generation.
- ii. Fossil fuel powered generators will continue to drive the off-grid power generation with annual growth rate of 3%.
- iii. Small hydro and solar will contribute 53 MW and 97 MW, respectively.
- iv. Grid generation remains poor at 10,000, 5000, 500, 400, 100, 100 MW, respectively, of gas, large hydro, solar PV utility, coal, wind, and biomass.
- v. Transmission losses will remain at 17%.

Table 1 shows the baseline's key assumptions used in the Nigeria LEAP model.

3.2.2 Current policy scenario (CPS)

The current policy scenario is one of the emissions reduction scenarios proposed in this study. This scenario models the pathway of existing national energy policies such as the Energy Transition Plan (ETP) and the Nationally Determined Contribution (NDC). Thus, assumptions and input were made based on the ETP, the NDC, the National Renewable Energy and Energy Efficiency Program. The CPS differs from the baseline scenario in that the baseline continues on the historic and current trend of poor power generation, energy mix and emissions, while the CPS imagines a future trajectory that is driven by existing government energy policies. Thus, like the ETP, about 90% of power generation in this scenario is attributed to renewable energy; electric vehicles is expected to replaced more than 85% of fossil fuel-powered cars and buses by 2060; and efficient household technologies are to replace 85% of inefficient household technologies by 2060 (FGN 2021). In addition, the scenario assumes that backup fossil fuel generators would be eliminated by 2050. See Table 2 for detailed description of the power mix and key policy assumptions supporting this scenario.

Table 2 Technology penetration assumptions across scenarios

Supporting Programs	Detailed Consideration		
	Current Policy Scenario (Blend of NDC and ETP)	Gas Economy Scenario	Renewable Energy Scenario
National Energy Policy (FGN 2022).	By 2050, grid power mix consist of CSP (5000 MW), nuclear (8000 MW), large hydro (24000 MW), solar PV (150000 MW), wind (20000 MW), biomass (9500 MW).	Grid power consists of CSP (1000 MW), Nuclear (15000 MW), Solar PV utility (45000 MW), Wind (1000 MW), biogas from waste (5000 MW), Biomass (1000 MW), coal (0 MW), large hydro (24000 MW), Gas (15000 MW), gas with CCS (130000 MW)	Grid power consists of CSP (15000 MW), Nuclear (20000 MW), Solar PV utility (45000 MW), Wind (25000 MW), biogas from waste (15000 MW), Biomass (20000 MW), coal (0 MW), large hydro (24000 MW), Gas (0 MW)
Mambilla Hydro Project. (3050 MW)	Off grid power mix will be minigrid (10000 MW), standalone (10000 MW), rooftop PV (10000 MW), small hydro (3500 MW).	Off grid power consists of Hybrid solar minigrid (2500 MW), Rooftop PV (5000 MW), Standalone PV (2000 MW), small hydro (3500 MW)	Off grid power consists of Hybrid solar minigrid (27950 MW), Rooftop PV (27500 MW), Standalone PV (27500 MW), small hydro (3500 MW)
Energy for all: Solar power strategy (5 million SHS). Sustainable Energy for All Action Agenda. Nigeria Energy Transition Plan (FGN 2021)	Total power generation capacity by 2050 is 250 GW (share of grid 86.6%) T&D losses: 2030 (8%), 2060 (5%).	Carbon capture can capture 80% of carbon emissions by 2030, and 100% carbon emissions by 2060. Total power generation capacity by 2060 is 250 GW (58% from Gas CCS)	T&D losses: 2030 (8%), 2060 (5%). Total power generation capacity by 2060 is 250 GW (100% renewable with nuclear energy).

3.2.3 Gas economy scenario (GES)

The rationale for this scenario is that Nigeria will exploit its abundant gas reserves. In this scenario, it was imagined that Nigeria would aggressively utilise its large gas reserves for power generation in such a way that it can achieve universal energy access. This scenario assumes that by 2060, about 60% of power generation would be from natural gas and that natural gas power plant would be fully integrated with carbon capture technology. Also, the scenario assumes that 45% of vehicles would be electric driven and inefficient household technologies would be fully replaced by efficient technologies by 2060. The logic for this is that efficient end-use technology will help to reduce the energy demand, thus, supporting the reduction of emissions of gas-powered plant. See Table 2 for detailed description of the power mix and key policy assumptions supporting this scenario.

3.2.4 Renewable energy scenario (RES)

This scenario envisages ambitious emission reduction to net-zero by 2060. The scenario assumes that by 2060, the power technologies would be mostly renewable energy, with nuclear power plant making up for the shortfall, and that end-use technologies would operate more efficiently. Also, this scenario assumes that most sectors of the economy will be electrified and that the services sector (which comprises of the telecommunication industry, tourism, health care, education, hotels and accommodation, etc.) would experience a significant transformation, at least twice the performance of the services sector in the CPS. See

Table 2 for detailed description of the power mix and key policy assumptions supporting the RES scenario.

In Table 2, the data setting for all three scenarios pegged at 250 GW was guided by the ETP, while T&D losses was guided by the NDC. The energy mix of the CPS was guided by both the ETP and NDC. The energy mix for the GES and RES were guided by the national energy policy, Mambilla hydro project, and FGN (2023). Also, the energy mix of Table 2 are exogenous capacity; however, the LEAP model was constrained to meet any unmet power requirement endogenously using Eq. (2) - any excess requirement would be visible in the results section. Furthermore, LEAP employs its interpolation capability to determine the annual capacities of the power plants from the base year values and the end year values of Tables 1 and 2.

4 Results and discussions

The results of the Nigeria's net-zero dedicated modelling exercise in respect of the power sector are presented. The results include the power requirements, the energy generation, energy demand, annual power installation, and emissions, across different scenarios including, baseline, CPS, GES, and RES. The results are discussed by comparing the findings of the emission reduction scenarios with the baseline, and also, with the literature.

4.1 Results

Figure 2 is the capacity requirement of the grid and off-grid power generation across the scenarios up to 2060. As shown in Fig. 2, the baseline scenario suggests that power generation capacity will increase from 32 GW in 2018 to 139 GW by 2060. Figure 2 indicates that most of the power supply in the baseline scenario will be provided through off-grid electricity,

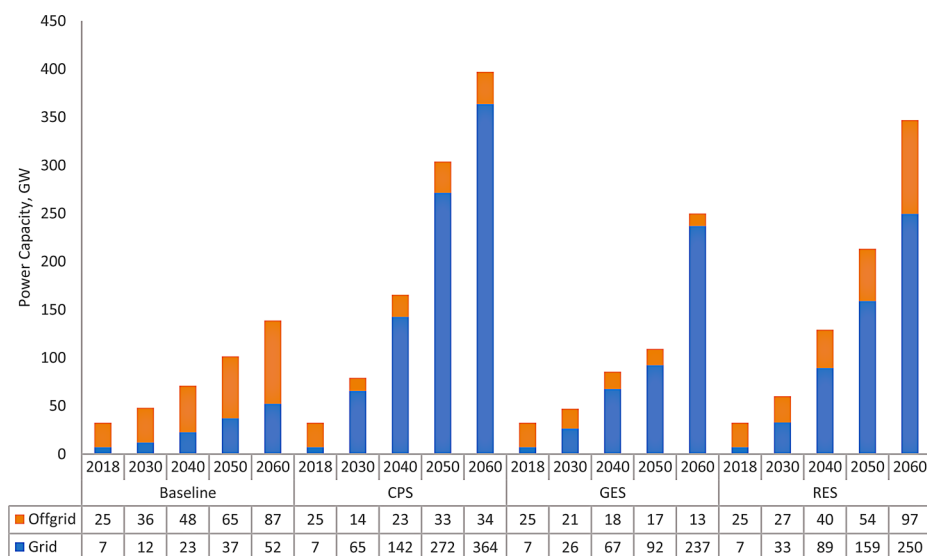


Fig. 2 Capacity of grid and off-grid power generation

contributing about 63% of electricity supply by 2060, from 78% contribution in 2018. The off-grid power supply is mostly from fossil fuel powered generators (see Fig. 3). In Fig. 2, the CPS scenario suggests that the total power capacity would be 79 GW, 165 GW, 304 GW and 398 GW, respectively, by 2030, 2040, 2050, and 2060. The figure suggests that a large portion of the CPS power capacity is expected from the grid power, about 91% of the total power capacity by 2060. This is consistent with the Nigeria Energy Transition (ETP) plan. Nevertheless, the total power capacity of 304 GW in 2050 exceeds the value of 250 GW specified in Table 2. The excess power is generated endogenously in LEAP using Eq. (2), since LEAP was constrained to rollout power plants to meet unmet capacities. In the GES, total power capacity is expected to be 47 GW, 85 GW, 109 GW, and 250 GW, respectively, by 2030, 2040, 2050, and 2060. Like the CPS, the power capacity level in the GES will be contributed mostly by the grid, about 95% of power capacity by 2060. The almost exclusive contribution of the grid to total power in the GES is partially due to the exploitation of the country's natural gas reserve in gas turbine power plants. In the RES, total power capac-

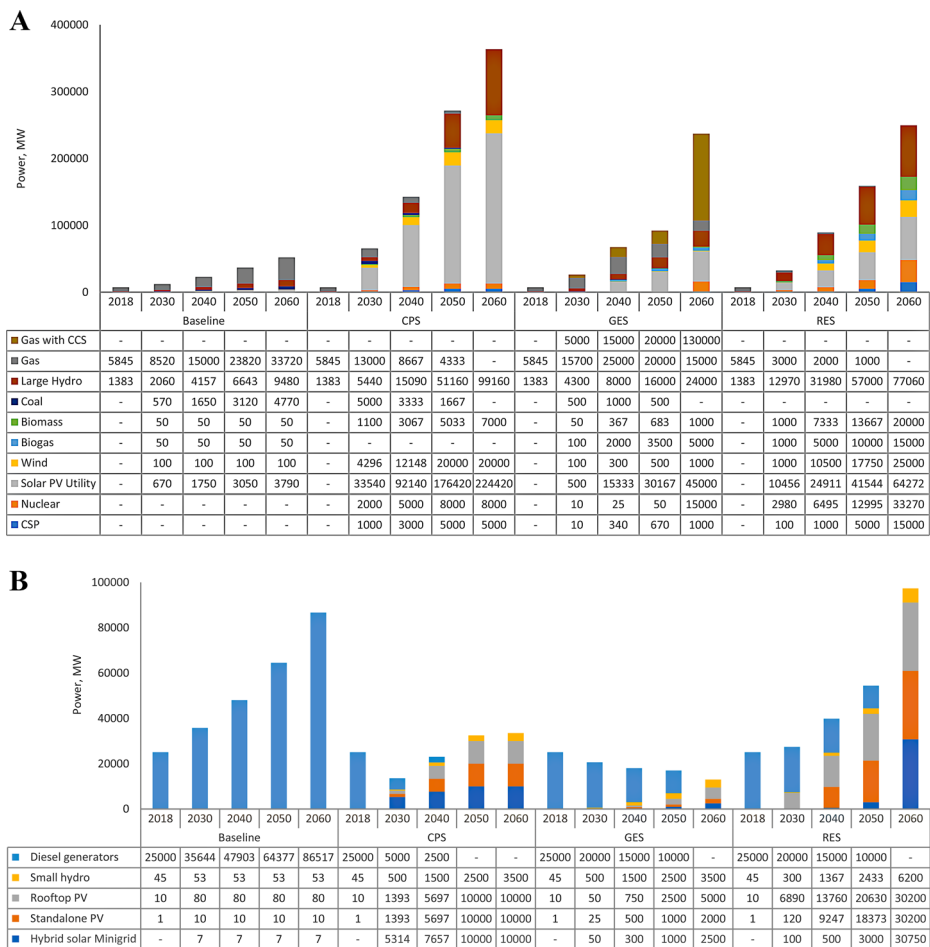


Fig. 3 a: Grid power generation capacity by technology. **b:** Off-grid power generation capacity by technology

ity is expected to be 60 GW, 129 GW, 213 GW, and 347 GW, by 2030, 2040, 2050, 2060, respectively. In the RES scenario, less than three-quarters of the power capacity by 2060 is expected from the grid power, contrary to the CPS and GES scenarios. Whereas, only 8 and 5% of power capacity in the CPS and GES, respectively, are expected from the off-grid, the RES shows that not less than 27% of power will come from the off-grid, a fact that is supported by the distributed nature of renewable energy.

Figure 2 suggests an increase in the power requirements of the emission reduction scenarios from the baseline scenario by 186%, 79%, and 150%, respectively, in the CPS, GES, and RES, by 2060. This increase in power requirement is because most sectoral end-use transition to electrification in a decarbonisation situation as electricity is one of the most demanded clean energies, howbeit, the generation process may be a direct source of emissions.

Figure 3 shows the power contribution by different power technologies across scenarios. Figure 3a shows the power contributions by technologies to the grid, while Fig. 3b shows the power contributions by technologies to the off-grid. The grid power consists of CSP, nuclear, solar PV utility, wind, biogas, biomass, coal, large hydro, gas, and gas with CCS technologies. Figure 3a suggests that, in the baseline scenario, the grid power mix is dominated by gas power plant without CCS, reaching 35GW (65% of the grid capacity) by 2060, and total capacity of renewable energy, except large hydro is about 7.6%, by 2060. The contributions of coal and large hydro are 9.2 and 18.2%, respectively. In the CPS, solar PV will contribute about one-half of the grid power mix at 42.4%, 55.8%, 58% and 56.5%, respectively, by 2030, 2040, 2050, and 2060. This is followed by large hydro. In the CPS, all gas and coal power plants would be retired by 2060, consistent with the Nigeria ETP. In Fig. 3a, the GES scenario suggests that most of the grid power capacity is gas power plants, contributing 79%, 59%, 43%, and 61%, respectively, by 2030, 2040, 2050, and 2060. The rebound in the share of the gas power plant in the grid power beyond 2050 is because of the envisaged maturity of CCS technology and that by 2060, all gas power plants in Nigeria would be integrated with CCS technology. In the RES of Fig. 3a, different power technologies contribute significantly to the grid power mix, unlike the CPS and GES that are tilted in favour of solar and gas, respectively. In other words, RES envisages a transition with greater diversity of clean energy in the Nigerian energy mix by 2060 compared to the other scenarios. However, large hydro and solar PV utility contribute more to the grid power at 77GW and 64 GW, respectively, by 2060. This is followed by nuclear, wind, biomass and CSP. The contribution by gas is almost insignificant, at 1GW (0.4%), by 2060.

Figure 3b shows the power contributions of different power technologies to the off-grid. The off-grid power technologies include hybrid solar minigrid, standalone PV (20W to 200 W), rooftop PV (200 W to 20 kW), small hydro and diesel generator. In the baseline scenario of Fig. 3b, diesel generator contributes most of the off-grid power at more than 99% of total off-grid power in 2030, 2040, 2050, and 2060, while the absolute contribution of off-grid renewables (all renewables) in the baseline scenario is only 150 MW by 2060. As previously noted, the contributions of modern off-grid renewable to Nigeria's power mix is currently 0.2% of installed capacity. The projected baseline situation in 2060 reflects a continued slow- uptake of renewables.

In the CPS scenario, there is gradual increase in the contributions of renewable energy technologies, with the contributions of hybrid solar minigrid, standalone PV, rooftop PV, and small hydro reaching 10 GW, 10 GW, 10 GW, and 3.5 GW, respectively, by 2060.

While the contribution of diesel generator decreases from 25 GW in 2018 to 0 GW by 2050, owing to the total phasing out of diesel generators. Similar observation is seen in the GES as is the case with the CPS, however, the penetrations of the renewable energy technologies is lower at 2.5GW, 2.0 GW, 5.0 GW, and 3.5 GW of the hybrid solar minigrid, standalone PV, rooftop PV, and small hydro, respectively, by 2060. The lower penetration than in the CPS is due to the assumption that there will be large gas exploration and utilization (for grid power generation) with some but limited penetration of renewable energy. In GES, diesel generators are phased out like in the CPS, however, the phase out of diesel generators is delayed until 2060.

In the off-grid power of RES scenario, there is higher penetration of renewable energy technologies than in the off-grid of CPS and the GES scenarios, with hybrid solar minigrid, standalone PV, rooftop PV, and small hydro reaching 30.8 GW, 30.2GW, 30.2GW and 6.2 GW, respectively, by 2060. This level of penetration is supported by the distributed nature of renewable energy, making it very suitable for off-grid generation, a factor that was not considered in the Nigeria ETP which is the backbone of the CPS scenario. Similar to the GES, the total phase out of the fossil fuel generators is delayed until 2060 to allow for a gradual stabilization and to also meet the share of power in the total energy demand. The absence of geothermal, tidal and wave energy from the scenarios and modelling work is deliberate because the potentials of these renewable energy sources in Nigeria are sketchy.

Figure 4 shows the energy output by scenarios. In Fig. 4, the baseline scenario suggests that the electrical energy output is expected to increase from 101 PJ in 2018 to 887 PJ by 2060, with energy from the grid contributing 83% by 2060. Although the off-grid contributed more to the power capacity in Fig. 2 at 63%, the grid contributes more to the energy output, owing to the low energy conversion efficiency of the diesel generator. For the emissions mitigation scenarios, Fig. 4 suggests higher energy output at 2147PJ, 1409PJ, and 2724 PJ in the CPS, GES, and RES, respectively, by 2060, than in the baseline. Earlier in Fig. 2, it was suggested that the CPS required more power capacity than the RES, however, the energy output of the RES scenario in Fig. 4 is higher than that of the CPS scenario by

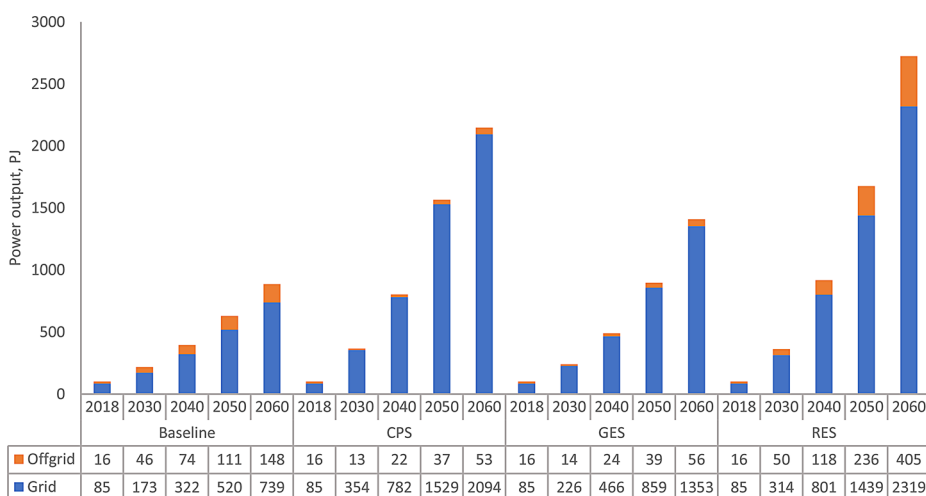


Fig. 4 Energy output of grid and off-grid generation

27%, in 2060, owing to the penetration of more efficient power technologies in the RES than in the CPS, nuclear power, for example.

In Fig. 5, the electrical energy demands of key energy sectors are presented across scenarios. Similar to the energy output, there is a gradual increase in the end-use energy in the baseline, reaching 744 PJ by 2060 from 84 PJ in 2018. In the CPS, GES, and RES, the end-use energy is expected to reach 2076PJ, 1312 PJ, and 2566 PJ, respectively, by 2060. This suggests that the electrical energy demand in the emission reduction scenarios will be 2.8, 1.8, and 3.4 times of the demand in the baseline scenario.

Figure 5 also suggests that apart from the baseline scenario, the emissions mitigation scenarios will experience a shift towards the electrification of the transport sector, with electrical energy use reaching 1165 PJ, 455 PJ and 1049 PJ, respectively, in the CPS, GES, and RES, by 2060. The figure also suggests that the services sector of the RES will demand more electrical energy than the services sector of the CPS and the GES at 79% and 115%, respectively, suggesting a boom in the activities of the services sector in the RES, for example, tourism, telecommunication, health care, etc. Although the mitigation scenarios have more use for electrical energy than the baseline, Fig. 5 suggests that the use of efficient end-use technologies such as energy savings lighting and inverter-type AC in the mitigation scenarios make the demand for electrical energy in the residential sector almost similar across all scenarios, at 348 PJ, 314 PJ, 305 PJ, and 386 PJ, in baseline, CPS, GES, and RES, respectively, by 2060.

Figure 6 shows the electricity per capita across the scenarios. In the baseline scenario of Fig. 6, there is a gradual increase in electricity per capita from 144 kWh in 2018 to 518 kWh by 2060, showing 260% increase from 2018 value. Due to improved electrification in the CPS, GES, and RES scenarios, electricity per capita increases by 142, 59, and 207% from the baseline value, respectively, by 2060, or 770, 470, and 1003% increase, respectively, from 2018 value. One would have expected higher values than what is shown in Fig. 6 for

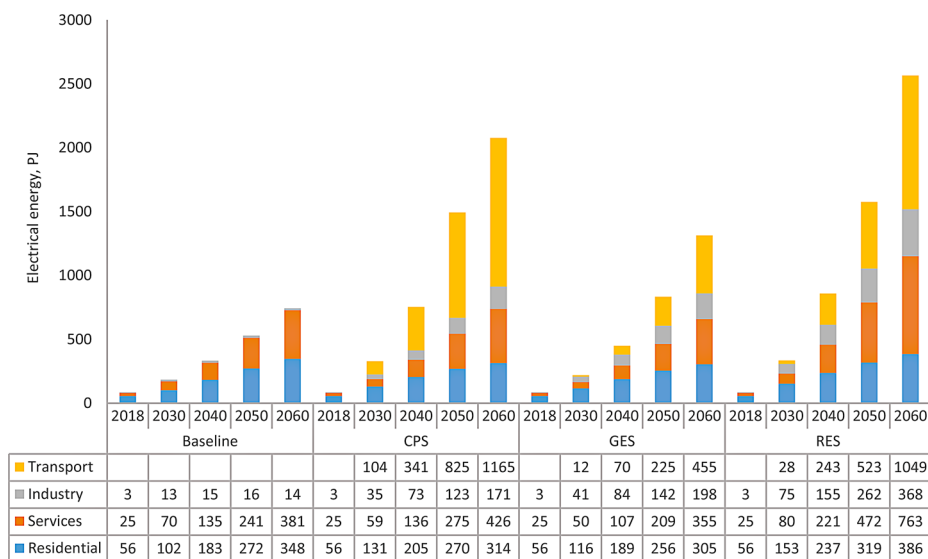


Fig. 5 Electrical energy demand by key sectors in the different scenarios

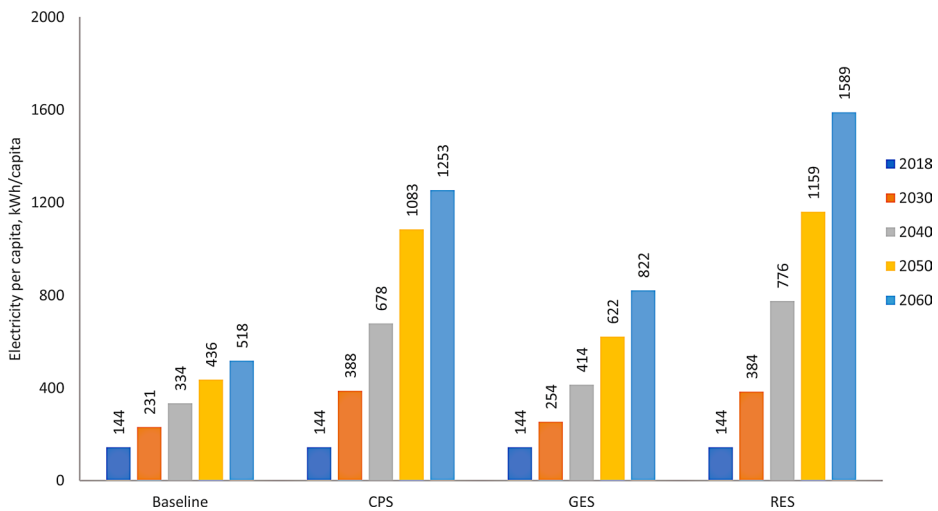


Fig. 6 Electricity per capita in the different scenarios

the emission reduction scenarios, however, end-use technologies are expected to be more efficient and energy saving.

Figure 7 shows the amount of power installations required, annually, by the different energy sources in the power mix across the scenarios. The baseline scenario suggests that 100 MW, 242 MW, 3 MW, 3 MW, 125 MW, 375 MW, and 1608 MW of solar, hydro, wind, biomass, coal, gas and diesel, respectively, are required, annually, to meet the 139 GW of power generation by 2060. This also suggests that 85.8% of total annual energy installations will be from fossil fuels, and only 24% from renewable energy. In Fig. 7, the emissions mitigation scenarios suggest that 6700, 1400 MW, and 4300 MW of solar are required to be installed in the CPS, GES, and RES, annually. The GES also requires that 3625 MW of gas power plants be installed, annually, which is about 58% of the total annual installation of the GES. Additionally, the RES also requires that 2056 MW, 875 MW, 830 MW, and 625 MW of hydro, biomass, nuclear, and wind be installed, annually.

Figure 8 shows the emissions associated with the different scenarios. These emissions are reported based on the 100-Year global warming potential of carbon dioxide equivalent ($\text{CO}_2\text{-eq}$). Figure 8 suggests that the emissions of the baseline scenario will continue to increase from 10.9 MMT $\text{CO}_2\text{-eq}$ in 2018 to 112.5 MMT $\text{CO}_2\text{-eq}$ by 2060, which is more than ten times the base year value. However, the mitigation scenarios show different levels of increase in emissions, and thereafter decrease to 0.5, 0.1, and 0.1 MMT $\text{CO}_2\text{-eq}$ in CPS, GES, and RES, respectively, by 2060. The CPS increases gradually and peaks at 18.3 MMT $\text{CO}_2\text{-eq}$ by 2040, the GES peaks at 15.3 MMT $\text{CO}_2\text{-eq}$ by 2050, RES peaks at 9 MMT $\text{CO}_2\text{-eq}$ by 2050. Figure 8 also suggests that most of the emissions in the CPS are from the grid power plants, but the reverse is the case in the RES (more emissions in off-grid than grid). The off-grid emissions in the RES emanates from the fossil fuel (diesel) back-up generators in use prior to phase out by 2060. In the GES, the grid power plants contribute more emissions than the off-grid in the early years of the scenario, however, by 2040, both the grid and off-grid contribute about the same quantity of emissions. In the GES, the grid

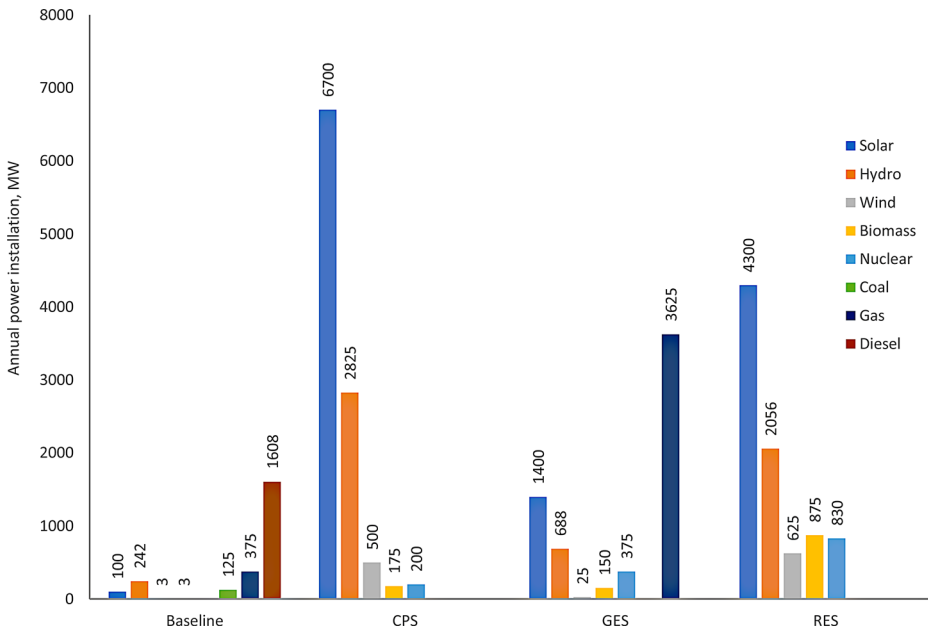


Fig. 7 Average annual power installation by energy sources across scenarios

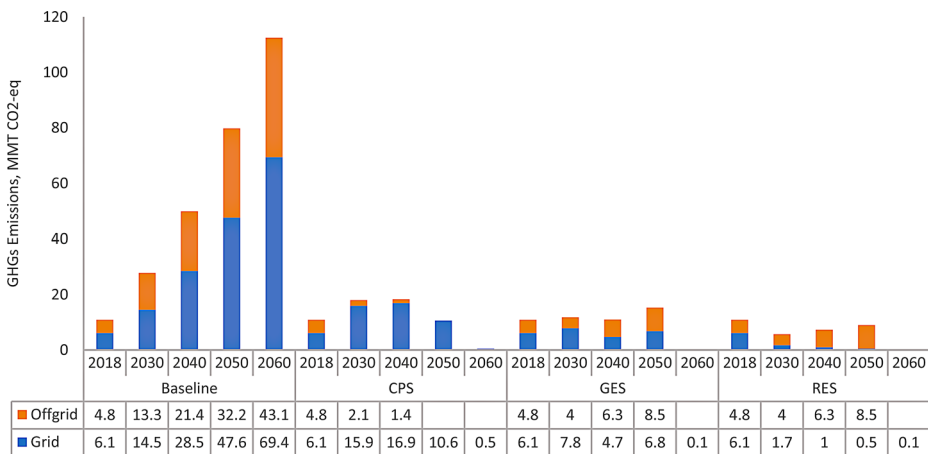


Fig. 8 GHGs Emissions of power generation in the different scenarios

emissions in the early years are due to low penetration of CCS technology, whereas, the off-grid emissions are due to fossil-fuel backup generators.

Figure 9 shows the emission factors across scenarios. In Fig. 9, the emissions factor of the baseline scenario increases from 388.5 tonne/GWh in 2018 to about 456.7 tonne/GWh by 2060. However, the CPS, GES, and RES show decrease in their emission factors from 386.3 tonne/GWh in 2018 to 0.8, 0.3, and 0.1 tonne/GWh, respectively, by 2060. The reduction in emission factors of the emission reduction scenarios is due to the lower emissions

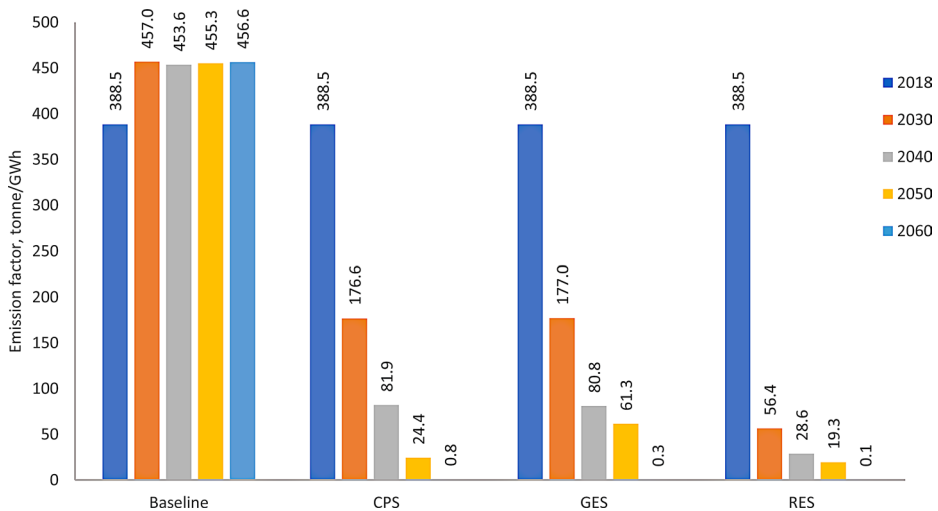


Fig. 9 Emission factors across scenarios

generated in these scenarios and their high energy outputs. The emission factor results also suggest a better performance of the RES than other scenarios in attaining net-zero.

Knowing that the power sector is only one aspect of the Nigerian economy, and that the RES strongly supports the electrification of other sectors, coupled with energy savings and efficient end-use technologies in the RES, it could be inferred that the RES scenario would perform better in attaining the country's net-zero ambition.

4.2 Discussions

The results obtained from the modelling exercise shows that credible pathways exist for Nigeria to meet its energy for all agenda while also decarbonising the power sector in pursuit of its net-zero ambition. Despite higher capacities and energy demand in the three emission reduction scenarios by 2060, the power sector's emissions could be significantly reduced to 0.5, 0.1, and 0.1 million metric tonne (MMT) $\text{CO}_2\text{-eq}$, respectively, in the CPS, GES, and RES, from 112.5 MMT $\text{CO}_2\text{-eq}$ in the baseline scenario in 2060. This finding suggests that the power sector could contribute significantly to the net-zero ambition of Nigeria.

The need to decarbonise the Nigeria power sector is very critical. It goes beyond emission reduction, and embraces clean energy access and electrification of other aspects of the economy. The results of the model have shown that an aggressive development, deployment and utilization of renewable energy should be a central element in Nigeria's decarbonisation strategy. Presently, available data suggests that the total renewable energy installation in Nigeria is not more than 3000 MW (IRENA 2023). In 2022, only 4 MW of renewable energy (from solar energy) was added to the total renewable energy installation (IRENA 2023). Practically, no installation was made for other renewable energy technologies. Till date, the total solar energy installed in Nigeria is still very poor, about 38 MW (Statistica 2023). Another source has shown that the contribution of solar energy to the total renewable energy portfolio is about 3% (IRENA

2023). This number is alarmingly low given the many years of relevant policy discussions and proclamations.

There are indications that a 100 MW solar project by Azura Independent Power Plant in Katsina state is under development, including another 200 MW Ashama solar project in Delta state (Owebor et al. 2021). While the completion of these two projects will represent significant progress, this capacity of solar energy installation in the country remains far below the expected annual installation needed to achieve the level of decarbonisation that is proposed in this study, 6700 MW, 1400 MW, and 4300 MW, annually, in CPS, GES, and RES, respectively. The slow pace of installation of solar energy in Nigeria, despite an ambitious NDC, ETP and the potentials of renewable energy sources in the country is a serious concern. The situation points to the need for radical policy and governance changes to translate proclamation into action.

Just like the case of solar, wind energy installation in Nigeria still remains very poor. The total installed capacity of wind energy in the country is the 10 MW wind farm in Katsina state, but, there is an ongoing 100 MW wind development in Plateau state (Owebor et al. 2021). The capacity of wind installation is far from achieving the 500 MW and 625 MW of annual wind installation needed in the CPS and the RES, respectively. Regarding biomass energy, the present installed capacity in Nigeria is not very clear, but (IRENA 2023) has shown that biomass contributes not more than 1% of the total renewable energy to power generation. In the records, hydro energy (mainly, large hydro power plant) is the only renewable energy in the country that is installed in large amount, roughly at 2010 MW (FGN 2022), contributing almost 96% of the renewable energy power generation in Nigeria (IRENA 2023). Also, there are ongoing large hydro power projects in Zungeru and Mambilla with capacities of 700 and 3050 MW, respectively (FGN 2022). Even at this development of hydro energy, it has been suggested that the capacity of hydro power (small and large hydro) in Nigeria is still underutilized (FGN 2022), and more efforts are still needed to meet the annual installation of 2825 and 2056 MW in the CPS, and RES, respectively, in this study. Also, Nigeria, in its policy document has demonstrated a strong interest to develop its nuclear energy for power generation (FGN 2022). But presently, there is no existing or known nuclear power development in the country. Yet, this study has shown that 830 MW of nuclear energy is needed annually, to support the base load electricity demand in the RES.

The results of this modelling demonstrates that so much efforts are still needed in investment in renewable energy, especially, solar, in the country for Nigeria to stand a chance of meeting its energy and climate goals. One key option to address the question of finance, often seen as a major barrier to energy investment in Nigeria, is to allocate a portion of the saving generated from the recent fossil fuel removal to a dedicated fund that can then be utilised to support investments and initiatives in renewable energy. An option previously proposed by the lead author suggests that Nigeria should allocate at least 20% of the savings from the subsidy removal regime specifically towards climate-related infrastructure and investments. Additionally, a portion of the funds can be allocated as catalytic funding to attract larger investments in renewable energy from global public and private entities. Based on estimated savings, this will release approximately, 16 billion naira annually for climate finance, which can be utilised to promote the sustainable development of Nigeria.

5 Conclusions and recommendations

Since July 2023, the world has been breaking records of all-time warmest month, at the same time, it is grappling with adverse climate change impact. Undoubtedly, there is dire need for deep decarbonisation necessary to attain the 1.5 °C thermal stability. In response to the call for countries to deeply decarbonise their economy, Nigeria took a commendable step by announcing its intention to attain net-zero by 2060 at COP26, held in Glasgow, UK. However, this pledge is not sufficient until it is backed by credible long-term strategies. In this work, a deep dive has been made into the decarbonisation of the Nigeria's power sector with focus on the net-zero pledge. This was done by investigating long-term model-based scenarios, which imagine the harnessing of the Nigeria gas project, utilization of the country's renewable energy resources, and harnessing the strengths of the Nigeria NDC and the Nigeria ETP in Gas Economy Scenario (GES), Renewable Energy Scenario (RES), and Current Policy Scenario (CPS), respectively. The scenarios were formulated taking into consideration key national policy programs, such as the solar power strategy, sustainable energy for all action agenda, and the national energy policy. The scenarios were modeled from 2018 to 2060 and implemented in the Low Emissions Analysis Platform. The findings suggested that the power sector's emission will continue to grow from 10.9 MMT CO_{2-eq} in 2018 to 112.5 MMT CO_{2-eq} by 2060 if the current developmental trend is continued without taking mitigation measures. However, if emission reduction paths are taken, emissions would drop to 0.5, 0.1, and 0.1 MMT CO_{2-eq} in the CPS, GES, and RES, respectively, by 2060. Additionally, the emissions reduction scenarios suggested that the power sector will play an important role in the decarbonisation of other sectors of the economy by contributing 2300, 1312, and 2566 PJ to the energy demand in the CPS, GES, and RES, respectively, by 2060, as against 744 PJ in the no-policy measures scenario, in the same year. In order to achieve the decarbonisation levels seen in the mitigation scenarios, it was shown that by 2060, the power generation capacity would be 398 GW, 250 GW, and 347 GW, in the CPS, GES, and RES, respectively.

From the findings of this work, it is suggested that government policies in the energy sector should give considerable attention to the power subsector, seeing that it can contribute its own quota to the Nigeria net-zero vision, at the same time, solve the energy access challenge in the country. Going by the annual power installation presented in this research, the present level of renewable energy penetration is not sufficient to drive a robust decarbonisation that can meet the net-zero target. The findings show that so much investment need to be done, most especially, investment in renewable energy. Therefore, it is crucial for the government to scale-up its efforts to promote wide adoption of renewable energy in the country in both grid and off-grid solutions. This can be achieved by creating a favourable political environment for investors, offering tax incentives for renewable energy technologies, promoting investments in low emission technologies, maximizing bilateral support for low emission power generation, replacing the discontinued oil and gas subsidy with a renewable energy subsidy, and discouraging the use of fossil fuels without carbon capture storage. It has been suggested that one way to raise climate change finance is to allocate at least 20% of the savings from the recent subsidy removal regime specifically towards climate-related infrastructure and investments.

Author contributions Chukwumerije Okereke (C.O) secured the funding; C.O. and Kesiena Owebor (K.O) conceptualized the manuscript; K.O. and Ogheneruona E. Diemuodeke (O.E.D) developed the methodology; C.O., K.O., and O.E.D. developed the scenarios; K.O. and O.E.D. curated the data and carried out the LEAP modelling; K.O. analyzed the data; C.O., K.O. and O.E.D. validated the results; C.O., K.O., and O.E.D., wrote, reviewed and edited the manuscript.

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Data availability See <https://unfccc.int/documents/638193>.

Declarations

Ethics approval and consent to participate Not applicable.

Consent for publication Not applicable.

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